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Assessing the Effectiveness of Large Language Model Support for Generating GDPR ROPA Documentation

Master Thesis at Ulm University

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1 Introduction

1.1 Kontext

The General Data Protection Regulation (GDPR) requires organizations to maintain a Record of Processing Activities (ROPA), a formal document that outlines how personal data is collected, processed, stored, and shared. Creating and maintaining these ROPA documents is often a resource-intensive process, especially for small and medium-sized organizations that lack specialized legal or data-protection expertise. Previous research developed an application that leverages Large Language Models (LLMs) to support users in generating GDPR-compliant ROPA documentation. The application provides three distinct interaction modes, each offering different degrees of assistance from the LLM, from basic text suggestions to highly automated drafting of ROPA entries. This system aims to reduce the effort needed to create legally compliant documentation while maintaining transparency and user control. Although the application provides a functional framework for assisted ROPA generation, its effectiveness, usability, and impact on documentation quality have not yet been systematically evaluated. To ensure that such a system can be confidently integrated into practical compliance workflows, both qualitative and quantitative assessments are needed. These evaluations must determine whether users can efficiently and correctly create ROPA documents, how the varying levels of LLM support influence performance and trust, and whether the system produces outputs aligned with GDPR requirements.

1.2 Reserach Problem and Research Questions

The goal of this thesis is to conduct a comprehensive evaluation of the existing LLM-supported ROPA generation application. The work will focus on assessing how the different interaction modes affect user experience, efficiency, output quality, and perceived reliability.

RQ1:

- How do varying degrees of LLM support influence Novice users during ROPA document creation?
- How do novice users perceive the usability, efficiency, and satisfaction of the three interaction modes (Form, Ask, Chat)?
- How does cognitive load differ across the three modes for users with no prior GDPR knowledge?
- Which mode do novice users find most intuitive and prefer, and why?
- How confident do users feel about the quality of their generated documents in each mode? (This is now crucial - perception vs. reality)

RQ2:

- How do varying degrees of LLM support influence Document quality when used by novice users?
- How does document quality differ across the three generation modes based on semantic similarity to an expert-created reference (BERTScore)?
- Compare documents using an LLM
- Are all fields present? (Does my pipeline work)
- Is there a discrepancy between user confidence (RQ1) and document quality (RQ2)?

1.3 3. Teil: Objectives und Contributions

Thesis Organization

- **Chapter 1: Introduction**
 - Motivation, research questions, and contributions
- **Chapter 2: Background and Related Work**
 - GDPR and ROPA
 - Related work in AI-assisted legal document generation
- **Chapter 3: Methodology**
 - Brief overview of ROPAgen
 - User study design (within-subjects, balanced Latin Square)
 - Data collection instruments (SUS, R-TLX, confidence questions)
 - Document quality assessment (BERTScore, LLM-as-a-judge)
- **Chapter 5: Evaluation and Results**
 - RQ1: User experience and preferences
 - RQ2: Document quality and confidence-quality gap
- **Chapter 6: Discussion**
 - Interpretation of findings and implications
 - Limitations and comparison with related work
- **Chapter 7: Conclusion**
 - Summary of contributions and future work

2 Background

2.1 GDPR

The General Data Protection Regulation (GDPR), implemented on May 25, 2018, represents the most stringent and comprehensive privacy and security framework in the global digital landscape [3]. While originated and enacted by the European Union, the regulation maintains an extraterritorial scope, imposing legal obligations on any organization—regardless of its physical location—that collects or processes data belonging to individuals within the EU. This landmark legislation is rooted in the 1950 European Convention on Human Rights, which established privacy as a fundamental right. It serves as a necessary evolution of the 1995 European Data Protection Directive, modernizing legal protections to address the rapid proliferation of cloud services, social media, and large-scale data breaches that have characterized the early 21st century. [3, 2]

Under the GDPR, the definitions of roles and data types are expanded to capture the complexity of modern informatics. Personal data is broadly defined as any information relating to an identifiable individual, encompassing everything from names and email addresses to biometric data, location information, and web cookies. The regulation distinguishes between the data subject (the individual), the data controller (the entity determining the "why" and "how" of processing), and the data processor (third parties handling data on behalf of a controller). To ensure ethical handling, all processing must adhere to seven core principles: lawfulness, fairness and transparency; purpose limitation; data minimization; accuracy; storage limitation; integrity and confidentiality; and accountability. This last principle is particularly vital, as it shifts the burden of proof to the organization, requiring documented evidence of compliance. [3]

The regulation mandates that processing is only legal if justified by one of six spe-

cific bases, most notably through consent, which must be freely given, specific, and unambiguous. Other justifications include contractual necessity, legal obligations, or a legitimate interest that does not override the subject's rights. Furthermore, the GDPR introduces the concept of data protection by design and by default, requiring that privacy considerations be integrated into the initial development phases of any product or service. Large-scale processors or public authorities are often required to appoint a Data Protection Officer (DPO) to oversee these internal processes and serve as a liaison with regulatory bodies. [3]

For individuals, the GDPR grants a robust suite of privacy rights, including the right to be informed, the right of access, the right to rectification, and the right to erasure (also known as the "right to be forgotten"). Subjects also possess the right to data portability and protections against automated decision-making and profiling. To enforce these standards, the regulation possesses significant "teeth"; authorities can levy harsh penalties for non-compliance. Consequently, the GDPR has transformed data privacy from a secondary IT concern into a primary strategic and legal priority for enterprises worldwide. [3].

2.2 Records of Processing Activities under Article 30 GDPR

Article 30 of the General Data Protection Regulation (GDPR) establishes the obligation for controllers and, where applicable, processors to maintain a Record of Processing Activities (RoPA). The RoPA functions as a core accountability instrument within the GDPR framework, operationalizing the principle that organizations must not only comply with data protection requirements but also be able to demonstrate such compliance to supervisory authorities. Its primary purpose is to provide a structured, transparent, and up-to-date overview of how personal data are processed within an organization, thereby enabling effective oversight, risk assessment, and regulatory control. [2, 8]

From a regulatory perspective, the RoPA serves several interrelated objectives. First, it enhances organizational awareness of data processing operations by requiring a systematic mapping of purposes, data flows, and safeguards. Second,

it supports data protection by design and by default, as organizations must explicitly consider data categories, retention periods, and security measures. Third, it enables supervisory authorities to efficiently assess compliance, particularly in the context of audits, investigations, or following personal data breaches. As such, the RoPA is not merely a formal record, but a living compliance document that reflects actual processing practices. [2]

Article 30 distinguishes between the obligations of data controllers and those of data processors. Controllers are required to document processing activities carried out under their responsibility, while processors must maintain records of processing performed on behalf of controllers. These records must be maintained in writing, including in electronic form, and must be made available to the competent supervisory authority upon request. Although the GDPR provides a limited exemption for organizations employing fewer than 250 persons, this exemption does not apply where processing is non-occasional, involves special categories of personal data, concerns criminal convictions, or is likely to result in risks to the rights and freedoms of data subjects. Consequently, in practice, many small and medium-sized organizations remain subject to the RoPA requirement. [2]

The content of a RoPA is explicitly defined by Article 30 GDPR and is intended to be concise, structured, and easily comprehensible. Excessive or irrelevant information should be avoided, as clarity and usability are essential for both internal governance and external supervision. At a minimum, a RoPA should include the following elements.

- Name and contact details of the data controller (and, where applicable, joint controllers, representatives, and the data protection officer).
- Purposes of the processing activities.
- Categories of data subjects.
- Categories of personal data processed.
- Categories of recipients, including recipients in third countries or international organizations.
- Details of transfers of personal data to third countries or international organizations, including safeguards where applicable.

- Time limits for the erasure of different categories of personal data.
- A general description of the technical and organizational security measures implemented pursuant to Article 32 GDPR.

Where an organization acts as a data processor, the RoPA must additionally reflect the processing carried out on behalf of each controller, including the identity of those controllers, the categories of processing involved, relevant international data transfers, and applicable security measures. In both cases, the RoPA should be sufficiently detailed to provide a faithful representation of processing activities, while remaining accessible to non-technical stakeholders. [2]

The GDPR does not prescribe a fixed update cycle for RoPAs; instead, it requires that records remain accurate and up to date. Consequently, any substantive change in processing purposes, data categories, recipients, or safeguards should trigger a review and update of the RoPA. Many organizations delegate the responsibility for maintaining the RoPA to a data protection officer, where one is appointed, to ensure consistency, avoid duplication of effort, and reduce the risk of omissions. [2, 7]

Finally, Article 30 emphasizes the external relevance of RoPA. Organizations must be prepared to present their records to supervisory authorities upon request, and such requests may be accompanied by demands for supplementary documentation, including privacy notices, consent records, or contractual arrangements. In this sense, RoPA operates as a foundational compliance artifact, anchoring broader data protection governance and demonstrating an organization's commitment to accountability under the GDPR. [2, 7].

2.3 Related Work

"A Systematic Comparison Between Open- and Closed-Source Large Language Models in the Context of Generating GDPR-Compliant Data Categories for Processing Activity Records" (von Schwerin et al., 2024) [8]

- Domain Focus and Research Gap
 - The sources focus on the automation of GDPR compliance documentation, specifically the generation of data categories (e.g., contact info, payment data) for Records of Processing Activities (ROPA).

- While general legal AI research exists for tasks like case retrieval and judgment prediction, the authors identify a significant research gap regarding the generation of structured, regulatory-compliant outputs.
- Existing legal benchmarks, such as LawBench or LegalBench, primarily evaluate static knowledge recall rather than the ability to produce documents that adhere to the precise formatting and terminology required by the GDPR.
- Methodology and Evaluation
 - The study employs a quantitative experimental design comparing four open-source models (Qwen2-7B, Meta-Llama-3-8B, zephyr-7b-beta, and orca_mini_v7_7b) against the closed-source GPT-4 as a baseline.
 - The methodology is structured around four experiments: basic performance, model-specific prompting, few-shot prompting, and Retrieval-Augmented Generation (RAG).
 - To evaluate the documents, the authors developed a comprehensive scoring metric called a Model Utility Score.
 - This metric weights five key criteria:
 - * Latency (Weight 1): Response speed.
 - * Structure and Style (Weight 2): Adherence to list formats and precise language.
 - * Grammar (Weight 3): Credibility and professional standards.
 - * Validity (Weight 4): Relevance of data categories to GDPR guidelines.
 - * Contextual Understanding (Weight 5): Comprehensiveness and lack of redundancy in covering specific data fields.
 - Additionally, they utilized two specialized benchmarks: a multiple-choice benchmark for contextual knowledge and a generation benchmark to compare LLM outputs against a reference database of correct data categories.

- Key Results and Synthesis
 - The findings demonstrate that open-source models, particularly Qwen2-7B, can achieve performance comparable to or even exceeding GPT-4 in generating structured ROPA data categories.
 - Synthesis of the results suggests:
 - * Prompting Nuance: Surprisingly, model-specific prompts aligned with pretraining formats did not necessarily improve performance; simplicity and clarity in prompts often yielded better results.
 - * Context Augmentation: Few-shot prompting had a limited impact on quality, while RAG showed mixed results.
 - * RAG improved performance for categories explicitly found in the source text but caused models to struggle with novel categories and ignore structural instructions (like list formatting).
 - * Viability: The study validates that open-source LLMs are cost-effective and privacy-preserving alternatives for ROPA documentation, as they allow for local deployment to satisfy strict GDPR data handling requirements.
- Limitations
 - The sources highlight several critical limitations for AI-driven compliance:
 - Ambiguity: LLMs still struggle with ambiguous legal language (e.g., terms like "necessary" or "adequate") and variations in jurisdiction-specific terminology.
 - Static Knowledge: The models rely on static datasets and may not reflect real-time updates to evolving GDPR regulations.
 - Hallucination: A persistent risk remains where models generate plausible but legally incorrect information, necessitating a "human-in-the-loop" review process.
 - Scope: The evaluation was limited specifically to data category generation, which may not generalize to more complex GDPR tasks like Privacy Impact Assessments.

"Structured Legal Document Generation in India: A Model-Agnostic Wrapper Approach with VidhikDastaavej" by Nigam et al. (2025) [6]

- Domain Focus and Research Gap
 - The study addresses the automation of private legal document drafting within the Indian legal system.
 - The authors identify a significant research gap: while previous work has extensively covered public legal tasks like judgment prediction and case summarization, the generation of private, structured legal documents remains largely unexplored due to data confidentiality and the complexity of legal drafting standards.
- The AI Generation Framework
 - The researchers developed NyayaShilp, a domain-adapted model based on LLaMA-2-7B, which underwent continued pretraining (CPT) on Indian case laws and supervised fine-tuning (SFT) on a specialized dataset.
 - Central to their approach is the Model-Agnostic Wrapper (MAW), a two-phase framework designed to improve long-form coherence:
 - * Phase 1 (Section Title Generation): The model generates a structured list of section titles based on user input, which the user can manually refine.
 - * Phase 2 (Section Content Generation): The system iteratively generates content for each section, utilizing a vector database (ChromaDB) to store and retrieve summaries of previous sections to maintain logical flow and reduce hallucinations.
 - * The system is integrated into an interactive Human-in-the-Loop (HITL) interface, allowing users to specify document types, customize sections, and refine AI-generated drafts.
- Methodology and Evaluation
 - To support this work, the authors introduced VidhikDastaavej, a novel, anonymized dataset of 489 private legal documents (e.g., petitions, affidavits, lease deeds).

- Their evaluation methodology is comprehensive and mirrors the dual-evaluation structure of your thesis:
 - * Quantitative Lexical and Semantic Metrics: The study uses standard metrics like ROUGE, BLEU, and METEOR, alongside BERTScore to measure semantic similarity between generated and reference texts.
 - * Automatic LLM-based Evaluation (LLM-as-a-judge): They employ G-Eval, a GPT-4-based framework, to assess coherence, factual accuracy, and completeness.
 - * Qualitative Expert Assessment: Three legal experts evaluated the documents on a 1-10 Likert scale for factual accuracy and completeness, with inter-annotator agreement (IAA) measured via Intraclass Correlation and Krippendorff's Alpha.
- Key Results and Synthesis
 - The findings indicate that direct fine-tuning (SFT) on small datasets can actually lead to performance degradation and hallucinations.
 - In contrast, the structured wrapper approach (MAW) significantly enhanced document quality, achieving scores comparable to GPT-4o even when using smaller open-source models.
 - Experts confirmed that the wrapper-assisted models produced more coherent, well-structured, and legally valid documents compared to base models.
- Limitations
 - Dataset Diversity: The training data was imbalanced, with a heavy focus on case judgments, leading to difficulties in generating underrepresented legal formats like contracts or agreements.
 - Hallucinations: While the wrapper reduced inconsistencies, it did not entirely eliminate hallucinations, suggesting a need for more robust fact-verification modules.
 - Real-World Usability Testing: The study primarily relied on expert evaluation; large-scale user feedback from practicing lawyers and broader

usability testing across different jurisdictions were identified as necessary future steps.

- Computational Constraints: The research did not fully explore larger architectures (e.g., LLaMA-3-70B) for fine-tuning due to resource limitations.

"LLM-assisted Extraction of Regulatory Requirements: A Case Study on the GDPR" (Abualhaija et al., 2025) [1]

- Domain Focus and Research Gap

- The paper addresses the knowledge gap between legal interpretation and software development, specifically focusing on the extraction of technical requirements from the General Data Protection Regulation (GDPR).
- While the GDPR establishes fundamental data subject rights, it often lacks the detailed technical guidance necessary for software implementation, leading to low compliance rates.
- The authors identify a specific research gap in the requirements engineering (RE) literature: most studies provide only generic mentions of the GDPR, with very few investigating specific data subject rights—such as the right of access (ACC) and the right to portability (PRT)—in-depth.
- Furthermore, the study explores the potential for Large Language Models (LLMs) to bridge this gap by leveraging additional legal sources (guidelines, case law, etc.) beyond the primary text of the GDPR.

- Methodology and Evaluation

- The authors utilized a mixed-methods approach involving both qualitative manual extraction and quantitative automated generation:
 - * Manual Baseline: In collaboration with six legal and RE experts, they manually extracted 108 reference requirements (61 for ACC; 47 for PRT) from mandatory legal sources, advisory guidelines, and academic literature.
 - * Automated Approach (XTRAREG): They developed an automated approach using GPT-3.5 and GPT-4o combined with Retrieval-Augmented Generation (RAG).

- * They tested different prompting techniques, including Zero-Shot Learning (ZSL) and Chain-of-Thought (CoT), to instruct the LLM to generate requirements informed by indexed legal documents.
- * Dual-Evaluation Framework:
 - Quantitative: The authors used automated NLP metrics to compare generated text to the manual ground truth, including precision/recall-oriented metrics (BLEU, ROUGE-L, METEOR) and the semantic similarity metric BERTScore.
 - Qualitative: Legal experts performed an extrinsic assessment of the generated requirements based on Correctness (factually accurate), Groundedness (valid legal references), Plausibility (reasonable rationale), and Coverage (percentage of reference requirements identified).
- Key Results and Synthesis
 - The study found that GPT-4o significantly outperformed GPT-3.5, particularly in the volume and completeness of generated requirements.
 - Key findings include:
 - * Accuracy and Correctness: XTRAREG achieved a correctness accuracy of 81.8% for ACC and 85.7% for PRT.
 - * Impact of RAG: The inclusion of RAG significantly improved performance across all metrics compared to "ablation" runs where the LLM relied only on its pre-trained knowledge.
 - * Semantic Similarity vs. Verbatim: The models achieved high BERTScores, indicating they successfully captured the semantic meaning of the requirements even when they did not match the manual baseline verbatim (as shown by lower BLEU scores).
 - * Source Preference: Despite being exposed to various sources via RAG, the LLMs showed a strong tendency to generate requirements predominantly from the GDPR itself, often overlooking more specific technical details found in supplementary guidelines.
- Limitations

- The authors highlighted several limitations inherent in using LLMs for regulatory tasks:
 - * Low Coverage: While the requirements generated were often correct, the LLM only covered a fraction of the total requirements identified by human experts.
 - * Perspective Errors: The LLM frequently adopted an incorrect perspective, writing from the viewpoint of the data subject (e.g., "the user may...") rather than the system (e.g., "the system shall...").
 - * Genericity and Actionability: Some outputs were non-actionable, providing verbatim legal excerpts rather than technical requirements, or failing to account for legal exceptions.
 - * Groundedness Issues: The models occasionally suggested incomplete or misgrounded references, sometimes citing recitals (which are non-binding in civil law systems) instead of specific normative articles

"Legal Documents Drafting with Fine-Tuned Pre-Trained Large Language Model" (Lin and Cheng, 2024) [5]

- Domain Focus and Research Gap
 - The paper focuses on the automated drafting of legal documents, specifically targeting "criminal facts" descriptions in Traditional Chinese fraud case verdicts.
 - The authors identify a significant research gap in the legal domain: traditional Natural Language Processing (NLP) methods rely heavily on manually annotated datasets, which are difficult and costly to obtain for specialized legal tasks.
 - Furthermore, most existing Large Language Models (LLMs) are trained on general text, leading to suboptimal performance when generating documents that require specific legal terminology and rigid formatting.
 - A critical secondary gap is the privacy and security risk associated with third-party AI systems; legal professionals are often reluctant to use cloud-based LLMs because they handle sensitive information.

- Methodology and Evaluation

- The researchers employed a quantitative approach to model training and a structural qualitative-to-quantitative metric for document evaluation:
- Local Fine-tuning: To address privacy concerns, the authors fine-tuned an open-source BLOOM-560M model on a local machine (RTX 3090).
- Dataset: They utilized a large, unlabeled dataset of 74,823 fraud verdicts from the Taiwan Judicial Yuan (2011–2021) without requiring Chinese word segmentation, thereby lowering the threshold for processing Chinese legal text.
- Evaluation Metric (Structural Correctness): A core contribution of the methodology is a decomposition method based on legal constitutive elements.
- The authors defined a specific order for document elements—Subject of Crime (LEO_SOC), Subjective Legal Element (LEO_SLE), Act (LEO_ACT), Victim (LEO_VIC), Causation (LEO_CAU), and Result of Hazard (LEO_ROH)—to evaluate if the LLM-generated draft follows the correct legal logic and format.
- Performance Metrics: The model was evaluated using perplexity and training/validation loss.

- Key Results and Synthesis

- The study demonstrates that local fine-tuning of a small-scale LLM (560M parameters) significantly improves its ability to generate legal drafts compared to the base model, which failed to produce relevant legal content.
 - * The fine-tuned model successfully generated drafts that adhered to the pre-defined structural order of legal elements (Equation 1), confirming the feasibility of using LLMs for specialized legal drafting.
 - * Synthesizing these results, the paper shows that domain-specific performance can be achieved without massive computational resources or cloud-based infrastructure, making it a viable solution for privacy-sensitive legal applications.

- Limitations The authors acknowledge several limitations:
 - Dataset Scope: The experimental dataset is relatively small and focused only on fraud cases, requiring broader validation.
 - Evaluation Metrics: Perplexity alone may not fully reflect the practical utility or quality of the documents in a real-world application.
 - Hardware Constraints: Fine-tuning larger models (e.g., 1.1B parameters) requires substantial GPU memory beyond standard consumer-grade hardware.
 - Hallucinations: The model occasionally produces irrelevant or non-existent terms (e.g., the word "nueve"), a persistent challenge in natural language generation.

"AI-Powered Legal Documentation Assistant" (Vayadande et al., 2024) [11]

- Domain Focus and Research Gap
 - The research focuses on the automation of legal document processing, specifically addressing document generation, comprehension, and abnormality detection.
 - The authors identify a significant research gap in the inherent complexity of legal systems, which traditionally rely on manual, time-consuming processes prone to human error.
 - Existing document processing systems are often characterized as inadequate because they cannot keep pace with the precision-oriented nature of legal documentation or the intricacies of specialized legal language.
- Methodology and Evaluation
 - The authors developed a system utilizing ChatGPT 3.5 integrated with OpenAI Embeddings and the LangChain framework,.
 - System Architecture: The methodology involves a multi-stage pipeline: data assembly of multilingual legal styles, data cleansing, and the use of FAISS vector stores for similarity searches.

- It employs PyPDF and Amazon Textract for document parsing and Named Entity Recognition (NER) combined with sentiment analysis to identify inconsistencies,.
 - Evaluation Approach: The evaluation is primarily quantitative, featuring a comparative analysis of the proposed LDA against existing market solutions like Kira Systems and Evisort.
 - Metrics used for evaluation include accuracy percentages, time efficiency (hours), and document processing speed (pages per minute).
 - Additionally, the study incorporates a qualitative element through a debugging phase that utilizes feedback and "balanced comments" from legal experts to refine the system's practical utility.
- Key Results and Synthesis
 - The study demonstrates that the AI-powered assistant significantly outperforms traditional methods:
 - * Accuracy: The system achieved an accuracy rate of 93–97%, which is comparable to or higher than established legal AI tools.
 - * Efficiency: It reduced the time required for document tasks from 20 hours (manual) to 5 hours (AI-assisted).
 - * Error Reduction: The error rate was lowered to 5%, compared to a 15% error rate in manual processing.
 - The synthesis of these results suggests that combining LLMs with specialized NLP toolkits (like NER and sentiment analysis) provides a "force of change" for making legal practices more frictionless.
- Limitations
 - Despite the high performance, the authors note several limitations:
 - * Accuracy Constraints: The system still lacks "absolute accuracy" and may struggle with the extreme nuances of certain judicial texts.
 - * Language and Format: While designed for multilingual potential, it does not yet apply to a wide enough variety of languages.

- * Furthermore, the accuracy of the system is highly dependent on the quality of input, such as the clarity of audio recordings or the success of digitizing handwritten notes.
- * Extraction Challenges: Using different text extraction tools for various formats (PDF vs. DOC) can lead to inconsistencies in how information is expressed.

"AI-Powered Interactive Legal Chatbot for the Department of Justice" (Surya et al., 2025) [10]

- Domain Focus and Research Gap Surya et al. (2025) focus on the Indian public legal assistance domain, specifically targeting the significant gap in accessibility to understandable and affordable legal information.
 - They identify a research gap in providing scalable, real-time, and multi-lingual support for citizens who struggle with complex legal jargon and the financial burden of professional counsel.
 - The work aims to democratize legal information, particularly for marginalized communities and those with limited resources.
- Methodology and Evaluation The methodology centers on the development of a three-tier conceptual and technical framework for an AI-powered chatbot.
 - The system architecture includes:
 - * Frontend: A JavaScript-powered single-page application for user interaction and document uploads.
 - * Backend: A Python Flask orchestrator handling API requests and language translation.
 - * LLM Service: LLaMA 3.1 accessed via the Ollama framework, utilizing specific prompt engineering for legal queries and document summarization.
 - Regarding evaluation, the paper employs qualitative illustrative use cases to demonstrate functionality rather than a rigorous quantitative user study.
 - These use cases simulate real-world scenarios, such as a user inquiring about consumer complaint procedures or another requesting a summary of a First Information Report (FIR).

- The evaluation also showcases multilingual interaction by processing a query in Hindi, translating it to English for the LLM, and returning the response in Hindi.
- Key Results and Synthesis The primary result is the successful integration of real-time query processing and document summarization within a unified legal assistant.
 - The authors synthesize these features to show how such a tool can provide 24/7 availability, reduce cost barriers, and simplify complex legal language for laypersons.
 - A significant contribution is the multilingual capability, which allows the system to reach a broader demographic by breaking down language barriers.
 - The system also emphasizes transparency through citation support, providing users with relevant legal references and government portal links to verify information.
- Limitations The paper identifies several critical limitations:
 - Accuracy and Hallucination: LLMs may generate plausible but incorrect information, which is particularly risky in the legal domain where precision is paramount.
 - Nuance and Context: The system may struggle with the subtleties of legal language, potentially leading to oversimplifications or misinterpretations in summaries of complex documents like FIRs or petitions.
 - Data Security: While the system handles sensitive personal data, current implementations require more robust measures like encryption and access controls to comply with India's Digital Personal Data Protection Act.
 - Ethical Concerns: There is an inherent risk of AI bias and a fine line between providing information and offering actionable legal advice, which users might misinterpret despite disclaimers.

"JuDGE: Benchmarking Judgment Document Generation for Chinese Legal System" (Su et al., 2025) [9]

- Domain Focus and Research Gap
 - The research focuses on the automated generation of complete legal judgment documents within the Chinese legal system, a task that demands complex legal reasoning and the integration of professional knowledge.
 - While previous work in Legal Judgment Prediction (LJP) primarily formulated the problem as a classification task—predicting discrete labels for charges or sentences—this paper addresses the generative gap by requiring models to produce structurally coherent and legally sound full-text documents.
 - The authors identify a critical research gap: the absence of standardized benchmarks and automated evaluation frameworks, which previously necessitated time-consuming and unscalable expert annotations.
- Methodology and Evaluation
 - The authors introduce JuDGE (Judgment Document Generation Evaluation), a benchmark designed to mirror the judicial syllogism process: using statutes and precedents as the major premise and case facts as the minor premise to reach a legal conclusion.
 - * Dataset: The benchmark includes 2,505 fact-judgment pairs from real criminal cases, supplemented by an external corpus of over 55,000 statutory articles and 103,000 past judgments.
 - * Quantitative Evaluation: The study employs a multi-dimensional automated framework to assess quality across four perspectives:
 - Penalty Accuracy: Measuring the precision of prison terms and fines using a normalized absolute difference score.
 - Convicting Accuracy: Utilizing Recall, Precision, and F1 scores to evaluate charge identification.
 - Referencing Accuracy: Assessing the correctness of statutory citations.
 - Documenting Similarity: Evaluating semantic alignment with ground truth via METEOR and BERTScore to capture nuanced legal

reasoning beyond simple lexical matching.

- Key Results and Synthesis

- The experiments compared several baselines, including direct generation, few-shot in-context learning (ICL), supervised fine-tuning (SFT), and Multi-source Retrieval-Augmented Generation (MRAG).
- Instruction-Following and Fine-Tuning: General and legal-domain LLMs performed poorly when used directly or via ICL, often failing to incorporate case-specific facts.
- However, SFT significantly improved performance across all models, making previously "unusable" legal LLMs competitive.
- RAG Effectiveness: The proposed MRAG approach, which retrieves both relevant statutes and similar cases, achieved the highest performance in convicting accuracy.
- Synthesis: While advanced techniques like RAG and SFT enhance document quality, current LLMs still struggle to consistently produce documents that match the legal rigor of professional ground truths, indicating that the task remains highly challenging.

- Limitations

- First, while MRAG improves recall for legal references, it often suffers from lower precision because current LLMs struggle to filter out irrelevant statutes provided by the retriever.
- Second, the automated metrics provided are currently focused on criminal law, which may limit their immediate applicability to other legal domains.
- Finally, the authors acknowledge that metrics like METEOR and BERTScore are proxies for semantic consistency and may not fully capture the absolute legal validity that a human expert would provide.

"CaseGen: A Benchmark for Multi-Stage Legal Case Documents Generation" (Li et al., 2025) [4]

- Domain Focus and Research Gap

- The authors focus on the Chinese legal domain, specifically the automation of drafting official judicial records, which traditionally requires significant manual effort and domain expertise.
- They identify a critical research gap: while LLMs show promise in legal applications, existing benchmarks are limited to general text tasks (like summarization) or simple, discriminative legal tasks (like judgment prediction or case retrieval).
- There was previously no benchmark capable of capturing the complexity, professional rigor, and multi-stage nature of drafting real-world legal documents.
- Methodology and Evaluation
 - CaseGen utilizes a dataset of 500 real-world legal cases annotated by five legal experts to ensure authenticity and completeness.
 - The methodology is defined by two core components:
 - * Multi-Stage Task Design: Instead of generating a document in a single step, the benchmark breaks the process into four distinct tasks: drafting defense statements, writing trial facts, composing legal reasoning, and generating judgment results.
 - * This approach aligns with the actual logic of legal writing and allows for a more nuanced assessment of an LLM's strengths and weaknesses.
 - * Evaluation Framework (Quantitative & Qualitative): The study employs a hybrid evaluation approach.
 - Quantitative: Traditional metrics like ROUGE-L and BERTScore are used but noted for their inability to capture logical coherence or factuality.
 - LLM-as-a-Judge: The primary evaluation is a pointwise scoring system (1-10) where an LLM (GPT-4o) acts as a judge.
 - This judge uses Chain-of-Thought (CoT) reasoning, task-oriented criteria (accuracy, completeness, logic), and reference-based comparisons to ground truth documents.

- Expert Validation: The effectiveness of the LLM judge was validated through human expert annotations, showing a 75% Spearman correlation, which suggests that LLM-as-a-judge is a reliable alternative for large-scale legal text evaluation.
- Key Results and Synthesis
 - General LLMs vs. Legal-Specific LLMs: Surprisingly, general-domain LLMs (such as GPT-4o and Qwen2.5) consistently outperformed legal-specific models like ChatLaw or LexiLaw.
 - The authors suggest that specialized models may suffer from limited base-model reasoning or information loss due to shorter context windows.
 - Current Performance Gap: Despite the capabilities of modern LLMs, most models achieved unsatisfactory scores (below 6 out of 10).
 - This indicates that existing AI still struggles with the logical rigor and complex reasoning required for professional legal drafting.
 - Metric Consistency: The study found that the LLM-as-a-judge score had significantly higher consistency with human experts than lexical matching metrics like ROUGE-L.
- Limitations
 - Geographic and Linguistic Scope: The benchmark is currently limited to the Chinese legal system, meaning the findings may not generalize to different legal frameworks or document formats.
 - Judge Reliability: While highly correlated with humans, LLM judges can still be influenced by inherent biases or adversarial inputs and cannot yet fully replace nuanced professional judgment.
 - Ethical Constraints: The study emphasizes that AI should remain a supportive tool for efficiency rather than a substitute for human legal expertise, especially given the risks of hallucinations in high-stakes environments.

3 Methodology

3.1 User Study

Study Design

- **Type:** Within-subjects experiment with counterbalanced order using a Balanced Latin Square design
- **Sample:** N = 73 university students (novice users, no prior GDPR experience)
- **Duration:** Approximately 90 minutes per participant
- **Counterbalancing:**
 - Balanced Latin Square with 6 groups (reflecting 3 AI modes tested in randomized sequences)
 - Each participant completes all three modes in a predetermined counter-balanced order
 - *Rationale:* Controls for order effects, learning effects, and fatigue

Task Scenario

- **Consistency:** Single standardized scenario used across all participants and modes
- **Content:** A relatable university-based data processing activity covering all GDPR Article 30 fields
- *Rationale:* Ensures comparability across conditions while remaining accessible to participants without legal expertise

- **LLM Model:** Mistral (consistent across all modes)

Participants

- **Recruitment:** N = 73 university students
- **Screening Criteria:**
 - Minimal to no GDPR knowledge (expected and required)
 - Prior AI tool usage experience (documented)
 - General technology comfort level (assessed)
 - No professional legal or compliance background
- **Onboarding:** Participants receive a brief standardized introduction to GDPR and ROPA concepts (e.g., short instructional video or text) before beginning the tasks
- **Reference Material:** Participants are shown an example of a high-quality ROPA document prior to starting

Data Collection

- **Quantitative Measures (collected after each mode):**
 - System Usability Scale (SUS): 10 standardized questions
 - Raw Task Load Index (R-TLX): 6 cognitive load dimensions
 - LLM-Specific Confidence Questions (7-point Likert scale):
 1. *Perceived Usefulness*: “The LLM was helpful in creating this document.”
 2. *Efficiency*: “This mode let me complete the ROPA faster than I could on my own.”
 3. *Transparency*: “I understood why the AI suggested certain GDPR fields.”

4. *Reliance Comfort*: “I felt comfortable relying on the AI’s suggested categories.”
 5. *Task-Specific Confidence*: “I am confident the AI identified the appropriate legal basis.”
 6. *Intention to Use*: “I would use this mode again for similar work.”
 7. *Professional Applicability*: “If I were responsible for GDPR in a real company, I would use this specific mode to generate my documentation.”
- Free-text qualitative feedback (collected after each mode)
 - Automatic completion time tracking (per mode)
- **Post-Experiment Ranking:**
 - Participants rank all three modes by overall preference after completing all conditions

Quality Assessment

- **BERTScore Evaluation:**
 - Compare all documents (73 participants × 3 modes) against an expert-created reference (ECR)
 - Aggregate scores by mode to establish baseline quality metrics
 - Perform pairwise comparisons between modes and against the ECR for comprehensive evaluation
- **LLM-as-a-Judge Validation:**
 - Evaluate a representative subset of mode-generated documents using Claude
 - Cross-validate BERTScore findings with LLM-based quality assessment
 - *Rationale*: Ensures BERTScore captures meaningful quality differences beyond lexical similarity

- **Confidence-Quality Gap Analysis:**

- Correlate self-reported confidence scores with actual document quality (BERTScore)
- Identify potential over-confidence or under-confidence patterns across modes

Statistical Analysis

- **RQ1 (User Experience & Preference):**

- Repeated measures ANOVA on SUS scores, R-TLX dimensions, and LLM confidence questions
- Post-hoc pairwise comparisons (with Bonferroni correction) to identify significant differences between modes
- Thematic analysis of free-text qualitative feedback and post-experiment rankings

- **RQ2 (Document Quality & Confidence Calibration):**

- Repeated measures ANOVA on BERTScore results across modes
- Correlation analysis between self-reported confidence and actual quality (BERTScore)

4 Evaluation and Results

5 Discussion

5.1 User Study Discussion

User Study Discussion

5.2 Document Discussion

Discussion on quality of generated document

6 Conclusion

A Quelltexte

In diesem Anhang sind einige wichtige Quelltexte aufgeführt.

```
1 public class Hello {  
2     public static void main(String[] args) {  
3         System.out.println("Hello World");  
4     }  
5 }
```

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Declaration

I hereby declare that I have written this thesis independently and have used no sources or aids other than those indicated.

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